

The Page File Size

 By default, page files are created with a starting size and a maximum size. These values allow Windows to resize the paging file as the system's need grows. It is more efficient to set an identical starting and max value so resources are not wasted by resizing a file.

To do this, choose 'Custom Size' for each page file, and set the initial and maximum sizes as identical.

As for what size to set them at, the best bet is to leave them at or slightly below the default 'Maximum' setting the system assigned, with a ceiling of 1 GB. This will be the amount of space reserved for the file, regardless of its current size. If you're creating multiple page files, split the amount between them.

Clean Up Unwanted Startup Programs

 Several programs run automatically when you turn on your system, and the number of such programs grows as you keep using your computer.

This leads to a gradual reduction in startup speed. Also, having programs that you only use selectively—or not at all—load automatically is a waste of resources.

To top it all, spyware, viruses and Trojans install themselves into the automatic start locations to ensure they run at startup.

To prevent the slowing down of your machine, you must take a look at the software that loads every time you start your PC, and disable what you don't need or can't identify.

The process is simple. Go to **Start Menu > Programs > Startup**. This is the directory XP uses to launch application shortcuts on bootup. If you remove the shortcuts from this directory, the apps will not load on startup.

This directory can also be a repository for spyware and viruses, so if there are files here that are not shortcuts and you don't recognise them, you should consider

removing them, as Windows will not place critical files in this directory.

The other way to remove unnecessary startup files is via the handy 'msconfig' utility. To use it, go to **Start > Run** and type in "msconfig". The 'Startup' tab in msconfig provides access to several other applications that start up at bootup and run in the background. By examining their filenames and directories, you should be able to get a feel of what is necessary and what is not.

Another method for removing programs from startup is through the programs themselves; many applications, such as MSN Messenger, offer the option of not starting at bootup.

Disable The Disk Performance Counter

 XP contains an inbuilt performance monitor that examines various areas of your system. This information can be called up using the performance monitor application, found in **Control Panel > Administrative Tools**. Most of us have little interest in this sort of performance statistics monitoring, that being more the territory of system administrators.

XP, however, is still monitoring away, and some of its observation tools use a considerable amount of resources. Disk monitoring is an example of this, and it's a good idea to turn the disk monitors off if you are not

planning on using the performance monitor app.

To disable the performance counter, go to **Start > Run**, type in "cmd", and type in "diskperf -N". This does away with the unnecessary monitoring that eats up valuable resources. To turn it back on, all you need to do is go back to the command prompt and type in "diskperf -Y".

Check And Set DMA Mode

 XP occasionally sets IDE hard drives and CD drives to the PIO transfer mode by default, which is slower than the standard DMA (Direct Memory Access) mode used by modern drives. To make sure your machine has not been slowed down, it makes sense to check your drive settings.

To check your drives, right-click 'My Computer' and select 'Properties', then 'Hardware' and then 'Device Manager'. Expand 'IDE ATA/ATAPI controllers'. Highlight 'Primary IDE channel' and click 'Properties'.

Go to the 'Advanced Settings' tab, and ensure the transfer mode is set to 'DMA if available'. Repeat these steps for the secondary IDE channel.

Mount A Hard Disk As A Folder In The C: Drive

 This tip works for any partition of any NTFS formatted drive except the partition with the Windows system files on it.

Windows XP, like Windows 2000 before it, allows you to 'mount' drives as folders in a pre-existing logical drive. So if your computer has a 20 GB disk formatted as a single partition and volume (drive C:), you could purchase a second drive, partition and format it through the Disk Manager, and then instead of giving it its own drive letter, add it to your C: drive as a directory.

Any files added to that directory would, of course, be stored on the new disk.

This can come in handy—certain applications such as databases are extremely large, but may not support storing data on a logically separate drive.

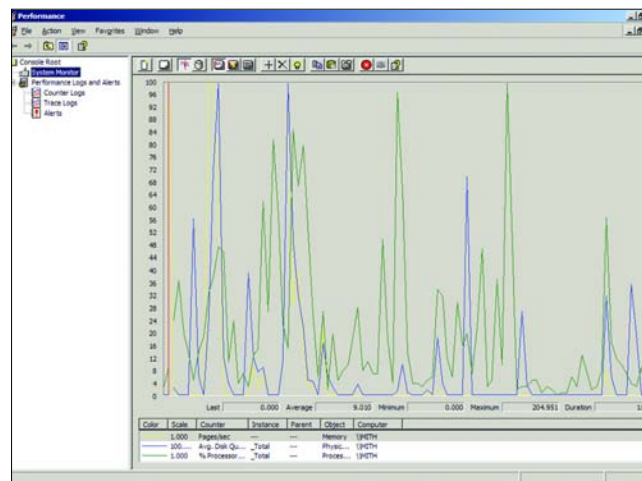
As far as Windows is concerned, a drive mounted as a directory is just a directory. This can also cut down on storage confusion and it's easy to do, though it can only be done with NTFS-formatted partitions.

Also, obviously, the boot partition cannot be used this way, although other partitions can be added to the boot partition.

If you have installed software on a partition you plan to mount as a directory, it is best to uninstall and reinstall it, since the move may stop the software from working correctly. Windows will warn you about this.

To mount a partition as a directory, go to **Control Panel > Administrative Tools > Computer Management > Storage > Disk Management**. Then right-click on the partition you wish to mount as a directory in the lower pane of the graphical partition window. Select 'Change drive letter and paths...' Remove the current option (if any), then click 'Add'. Choose 'Mount in the following empty NTFS folder', browse to the desired volume, and add a directory for your drive. Click 'OK'.

If you wish to return things to the way they were, simply repeat the



XP continuously monitors your hard disk performance!